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Inventor(s)	Noh; So-Young et al.

Display apparatus having a substrate hole

Abstract

A display apparatus can include a flexible substrate including a penetrating hole, a first thin film transistor including a first semiconductor layer, a second thin film transistor including a second semiconductor layer, and a first planarization layer on the first and second thin film transistors. The display apparatus can include a connection electrode on the first planarization layer electrically connected to the first or second thin film transistor, a second planarization layer on the first planarization layer, a first electrode on the second planarization layer electrically connected to the connection electrode, and a bank layer on the second planarization layer exposing a portion of the first electrode. The display apparatus includes a light-emitting layer on the portion of the first electrode exposed by the bank layer, the light-emitting layer including an emission material layer between first and second organic layers, and a second electrode on the light-emitting layer.

Inventors: **Noh; So-Young (Goyang-si, KR), Je; So-Yeon (Paju-si, KR), Kim; Ki-Tae (Seoul, KR), Moon; Kyeong-Ju (Paju-si, KR), Ji; Hyuk (Paju-si, KR)**

Applicant: **LG Display Co., Ltd.** (Seoul, KR)

Family ID: **1000008778507**

Assignee: **LG DISPLAY CO., LTD. (Seoul, KR)**

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Primary Examiner: Kim; Su C

Attorney, Agent or Firm: Birch, Stewart, Kolasch & Birch, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation application of U.S. patent application Ser. No. 16/943,889, filed on Jul. 30, 2020, which claims the priority benefit of Korean Patent Application No. 10-2019-0093100, filed in the Republic of Korea on Jul. 31, 2019, the entire contents of all these applications being hereby expressly incorporated by reference as if fully set forth herein into the present application.

BACKGROUND OF THE INVENTION
Field of the Invention
(1) The present invention relates to a display apparatus including a substrate hole which penetrates a device substrate.
Discussion of the Related Art
(2) Generally, an electronic appliance, such as a monitor, a TV, a laptop computer and a digital camera, includes a display apparatus to realize an image. For example, the display apparatus can include light-emitting devices. Each of the light-emitting devices can emit light displaying a specific color. For example, each of the light-emitting devices can include a light-emitting layer between a first electrode and a second electrode.
(3) A peripheral appliance such as a camera, a speaker and a sensor can be mounted in the display apparatus. For example, the display apparatus can include a substrate hole penetrating a device substrate which supports the light-emitting devices. The substrate hole can be disposed among the light-emitting devices. The peripheral appliance can be inserted into the substrate hole.
(4) However, in the display apparatus, external moisture can permeate through the substrate hole. The external moisture permeating through the substrate hole can move to the light-emitting device adjacent the substrate hole through the light-emitting layer. Thus, in the display apparatus, the light-emitting devices which are disposed adjacent the substrate hole may be damaged due to the external moisture permeating through the substrate hole.
SUMMARY OF THE INVENTION
(5) Accordingly, the present invention is directed to a display apparatus that substantially obviates one or more problems due to limitations and disadvantages of the related art.
(6) An object of the present invention is to provide an improved display apparatus capable of preventing damage of a light-emitting device due to external moisture permeating through a substrate hole.
(7) Another object of the present invention is to provide a display apparatus capable of simplifying a process for blocking the external moisture.
(8) Additional advantages, objects, and features of the invention will be set forth in part in the description which follows and in part will become apparent to those having ordinary skill in the art upon examination of the following or can be learned from practice of the invention. The objectives and other advantages of the invention can be realized and attained by the structure particularly pointed out in the written description and claims hereof as well as the appended drawings.
(9) To achieve these objects and other advantages and in accordance with the purpose of the invention, as embodied and broadly described herein, there is provided a display apparatus comprising a substrate hole penetrating a device substrate. A light-emitting device is disposed on the device substrate. The light-emitting device is spaced away from the substrate hole. The light-emitting device includes a first electrode, a light-emitting layer and a second electrode, which are sequentially stacked. At least one separating device is disposed between the substrate hole and the light-emitting device. The separating device includes at least one under-cut structure. A depth and a length of the under-cut structure are larger than a thickness of the light-emitting layer.
(10) The under-cut structure of the separating device can be in contact with the second electrode.

(11) The separating device can have a structure in which wide patterns and narrow patterns are repeatedly stacked. The narrow patterns can include a material different from the wide patterns.
(12) A width of each narrow pattern can be smaller than a width of each wide pattern. At least one of the narrow patterns can include a conductive material.
(13) A separation insulating layer can be disposed between the device substrate and the light-emitting device. A first thin film transistor can be disposed between the device substrate and the separation insulating layer. A second thin film transistor can be disposed between the separation insulating layer and the light-emitting device. An over-coat layer can be disposed between the second thin film transistor and the light-emitting device. Each of the first thin film transistor and the second thin film transistor can include a gate electrode, a source electrode and a drain electrode. Each of the narrow patterns can include the same material as one of the gate electrode, the source electrode and the drain electrode.
(14) A semiconductor pattern of the first thin film transistor can include a low-temperature poly-silicon (LTPS). A semiconductor pattern of the second thin film transistor can include an oxide semiconductor.
(15) Each of the first thin film transistor and the second thin film transistor can further include a gate insulating layer between the semiconductor pattern and the gate electrode, and an interlayer insulating layer between the gate electrode and the source and drain electrodes. Each of the wide patterns can include the same material as one of the separation insulating layer, the gate insulating layer, the interlayer insulating layer and the over-coat layer.
(16) A thickness of each wide pattern can be larger than a thickness of each narrow pattern.
(17) In another embodiment, the display apparatus comprises a device substrate. The device substrate includes a second region surrounded by a first region, and a third region disposed outside the second region. A substrate hole penetrates the first region of the device substrate. A light-emitting device is disposed on the third region of the device substrate. The light-emitting device includes a light-emitting layer between a first electrode and a second electrode. At least one separating device is disposed on the second region of the device substrate. The separating device has a structure in which a wide pattern and a narrow pattern are repeatedly stacked at least once. The narrow pattern has a smaller width than the wide pattern. A thickness of the narrow pattern is larger than a thickness of the wide pattern. A distance between a side surface of the wide pattern and a side surface of the narrow pattern is larger than a thickness of the light-emitting layer.
(18) The wide pattern and the narrow pattern can include an insulating material.
(19) The uppermost portion of the separating device can include the wide pattern.
(20) A first over-coat layer and a second over-coat layer can be sequentially stacked between the device substrate and the light-emitting device. The wide pattern disposed at the uppermost portion of the separating device can include the same material as the second over-coat layer.
(21) A thin film transistor can be disposed between the device substrate and the light-emitting device. A narrow pattern can include the same material as a gate insulating layer or an interlayer insulating layer of the thin film transistor.
(22) The narrow pattern can include silicon oxide based material.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The accompanying drawings, which are included to provide a further understanding of the invention and are incorporated in and constitute a part of this application, illustrate embodiment(s) of the invention and together with the description serve to explain the principle of the invention. In the drawings:
(2) FIG. **1** is a view schematically showing a display apparatus according to an embodiment of the

present invention;
(3) FIG. **2** is an enlarged view showing a peripheral of a substrate hole in the display apparatus according to the embodiment of the present invention;
(4) FIG. **3** is a view showing a cross-section of a pixel in the display apparatus according to the embodiment of the present invention;
(5) FIG. **4**A is a view showing a cross-section of a peripheral region of the substrate hole in the display apparatus according to the embodiment of the present invention;
(6) FIG. **4**B is an enlarged view of an area P**1** in FIG. **4**A;
(7) FIG. **5** is an enlarged view of an area K in FIG. **4**B; and
(8) FIGS. **6** to **15**B are views respectively showing a display apparatus according to other embodiments of the present invention, where FIG. **7**B is an enlarged view of an area P**2** in FIG. **7**A, FIG. **8**B is an enlarged view of an area P**3** in FIG. **8**A, FIG. **10**B is an enlarged view of an area P**4** in FIG. **10**A, FIG. **12**B is an enlarged view of an area P**5** in FIG. **12**A, FIG. **14**B is an enlarged view of an area P**6** in FIG. **14**A, and FIG. **15**B is an enlarged view of an area P**7** in FIG. **15**A.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(9) Hereinafter, details related to the above objects, technical configurations, and operational effects of the embodiments of the present invention will be clearly understood by the following detailed description with reference to the drawings, which illustrate some embodiments of the present invention. Here, the embodiments of the present invention are provided in order to allow the technical sprit of the present invention to be satisfactorily transferred to those skilled in the art, and thus the present invention can be embodied in other forms and is not limited to the embodiments described below.
(10) In addition, the same or extremely similar elements can be designated by the same reference numerals throughout the specification, and in the drawings, the lengths and thickness of layers and regions can be exaggerated for convenience. It will be understood that, when a first element is referred to as being “on” a second element, although the first element can be disposed on the second element so as to come into contact with the second element, a third element can be interposed between the first element and the second element.
(11) Here, terms such as “first” and “second” can be used to distinguish any one element with another element and may not define any order. However, the first element and the second element can be arbitrary named according to the convenience of those skilled in the art without departing the technical sprit of the present invention.
(12) The terms used in the specification of the present invention are merely used in order to describe particular embodiments, and are not intended to limit the scope of the present invention. For example, an element described in the singular form is intended to include a plurality of elements unless the context clearly indicates otherwise. In addition, in the specification of the present invention, it will be further understood that the terms “comprises” and “includes” specify the presence of stated features, integers, steps, operations, elements, components, and/or combinations thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or combinations.
(13) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example embodiments belong. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and should not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

Embodiments

(14) FIG. **1** is a view schematically showing a display apparatus according to an embodiment of the present invention. FIG. **2** is an enlarged view showing a peripheral of a substrate hole in the display apparatus according to the embodiment of the present invention. FIG. **3** is a view showing a cross-

section of a pixel in the display apparatus according to the embodiment of the present invention. FIG. **4**A is a view showing a cross-section of a peripheral region of the substrate hole in the display apparatus according to the embodiment of the present invention. FIG. **4**B is an enlarged view of an area P**1** in FIG. **4**A. FIG. **5** is an enlarged view of an area K in FIG. **4**B.

(15) Referring to FIGS. **1** to **5**, the display apparatus according to the embodiment of the present invention can include a device substrate **100**. The device substrate **100** can include an insulating material. For example, the device substrate **100** can include glass or plastic. The device substrate **100** can have a multi-layer structure. For example, the device substrate **100** can have a structure in which an insulating layer **102** is disposed between a first substrate layer **101** and a second substrate layer **103**. The second substrate layer **103** can include the same material as the first substrate layer **101**. For example, the first substrate layer **101** and the second substrate layer **103** can include plastic. The insulating layer **102** can include an insulating material.

(16) The device substrate **100** can include pixels PA defined by gate lines GL and data lines DL. A light-emitting device **500** can be disposed in each pixel PA. Each of the light-emitting devices **500** can emit light displaying a specific color. For example, each of the light-emitting devices **500** can include a first electrode **510**, a light-emitting layer **520** and a second electrode **530**, which are sequentially stacked.

(17) The first electrode **510** can include a conductive material. The first electrode **510** can include a metal having a relatively high reflectance. The first electrode **510** can have a multi-layer structure. For example, the first electrode **510** can have a structure in which a reflective electrode formed of a metal such as aluminum (Al) and silver (Ag) is disposed between transparent electrodes formed of a transparent conductive material such as ITO and IZO.

(18) The light-emitting layer **520** can generate light having luminance corresponding to a voltage difference between the first electrode **510** and the second electrode **530**. For example, the light-emitting layer **520** can include an emission material layer (EML) **522** having an emission material. The emission material can include an organic material, an inorganic material or a hybrid material. For example, the display apparatus according to the embodiment of the present invention can be an organic light-emitting display apparatus including the light-emitting layer **520** formed of an organic material.

(19) The light-emitting layer **520** can have a multi-layer structure in order to increase luminous efficacy. For example, the light-emitting layer **520** can include at least one first organic layer **521** between the first electrode **510** and the emission material layer **522**, and at least one second organic layer **523** between the emission material layer **522** and the second electrode **530**. The first organic layer **521** can include at least one of a hole injection layer (HIL) and a hole transporting layer (HTL). The second organic layer **523** can include at least one of an electron transporting layer (ETL) and an electrode injection layer (EIL). However, the disclosure is not limited thereto. For example, the first organic layer **521** can include at least one of the electron transporting layer (ETL) and the electrode injection layer (EIL), and the second organic layer **523** can include at least one of the hole injection layer (HIL) and the hole transporting layer (HTL).

(20) The second electrode **530** can include a conductive material. The second electrode **530** can include a material different from the first electrode. For example, the second electrode **530** can be a transparent electrode formed of a transparent conductive material, such as ITO and IZO. Thus, in the display apparatus according to the embodiment of the present invention, the light generated from the light-emitting layer **520** of each pixel PA can emit outside through the second electrode **530**.

(21) A driving current corresponding to a gate signal applied by a corresponding gate line GL and a data signal applied by a corresponding data line DL can be supplied to each light-emitting device **500**. For example, a driving circuit electrically connected to the corresponding light-emitting device **500** can be disposed in each pixel PA. The driving circuit can control the operation of the corresponding light-emitting device **500** according to the gate signal and the data signal. For

example, the driving circuit can include a first thin film transistor **200**, a second thin film transistor **300** and a storage capacitor **400**.
(22) The first thin film transistor **200** can include a first semiconductor pattern **210**, a first gate insulating layer **220**, a first gate electrode **230**, a first interlayer insulating layer **240**, a first source electrode **250** and a first drain electrode **260**.
(23) The first semiconductor pattern **210** can be disposed close to the device substrate **100**. The first semiconductor pattern **210** can include a semiconductor. For example, the first semiconductor pattern **210** can include a poly-silicon which is a poly crystalline semiconductor. The display apparatus according to the embodiment of the present invention is described that the first semiconductor pattern **210** includes a low-temperature poly silicon (LTPS).
(24) The first semiconductor pattern **210** can include a first source region, a first drain region and a first channel region. The first channel region can be disposed between the first source region and the first drain region. The first channel region can have a lower conductivity than the first source region and the first drain region. For example, the first source region and the first drain region can have a higher content of conductive impurities than the first channel region.
(25) The first gate insulating layer **220** can be disposed on the first semiconductor pattern **210**. The first gate insulating layer **220** can extend beyond the first semiconductor pattern **210**. The first gate insulating layer **220** can include an insulating material. For example, the first gate insulating layer **220** can include a silicon oxide based material (SiOx). The silicon oxide based material (SiOx) can include silicon dioxide (SiO.sub.2).
(26) The first gate electrode **230** can be disposed on the first gate insulating layer **220**. For example, the first gate electrode **230** can overlap the first channel region of the first semiconductor pattern **210**. The first gate electrode **230** can be insulated from the first semiconductor pattern **210** by the first gate insulating layer **220**. The first gate electrode **230** can include a conductive material. For example, the first gate electrode **230** can include a metal, such as aluminum (Al), chromium (Cr), copper (Cu), titanium (Ti), molybdenum (Mo) and tungsten (W).
(27) The first interlayer insulating layer **240** can be disposed on the first gate insulating layer **220** and the first gate electrode **230**. The first interlayer insulating layer **240** can extend along the first gate insulating layer **220**. The first interlayer insulating layer **240** can include an insulating material. The first interlayer insulating layer **240** can include a material different from the first gate insulating layer **220**. For example, the first interlayer insulating layer **240** can include silicon nitride based material (SiNx).
(28) The first source electrode **250** can be disposed on the first interlayer insulating layer **240**. The first source electrode **250** can be electrically connected to the first source region of the first semiconductor pattern **210**. For example, the first interlayer insulating layer **240** can include a first source contact hole partially exposing the first source region of the first semiconductor pattern **210**. The first source electrode **250** can include a portion overlapping with the first source region of the first semiconductor pattern **210**.
(29) The first source electrode **250** can include a conductive material. For example, the first source electrode **250** can include a metal, such as aluminum (Al), chromium (Cr), molybdenum (Mo), tungsten (W) and copper (Cu).
(30) The first drain electrode **260** can be disposed on the first interlayer insulating layer **240**. The first drain electrode **260** can be electrically connected to the first drain region of the first semiconductor pattern **210**. For example, the first interlayer insulating layer **240** can include a first drain contact hole partially exposing the first drain region of the first semiconductor pattern **210**. The first drain electrode **260** can include a portion overlapping with the first drain region of the first semiconductor pattern **210**.
(31) The first drain electrode **260** can include a conductive material. For example, the first drain electrode **260** can include a metal, such as aluminum (Al), chromium (Cr), molybdenum (Mo), tungsten (W) and copper (Cu). The first drain electrode **260** can include the same material as the

first source electrode **250**. For example, the first drain electrode **260** can be formed by the same process as the first source electrode **250**.

(32) The first source electrode **250** and the first drain electrode **260** can have a multi-layer structure. For example, each of the first source electrode **250** and the first drain electrode **260** can have a three-layer structure in which an intermediate layer being composed of a titanium (Ti) metal layer is disposed between a lower layer and an upper layer which are composed of an aluminum (Al) metal layer. The second thin film transistor **300** can be formed by a process different from the first thin film transistor **200**. For example, the second thin film transistor **300** can be disposed on a separation insulating layer **130** covering the first source electrode **250** and the first drain electrode **260** of the first thin film transistor **200**. The separation insulating layer **130** can include an insulating material. The separation insulating layer **130** can include a material different from the first interlayer insulating layer **240**. For example, the separation insulating layer **130** can include silicon oxide based material (SiOx).

(33) The second thin film transistor **300** can have the same structure as the first thin film transistor **200**. For example, the second thin film transistor **300** can include a second semiconductor pattern **310**, a second gate insulating layer **320**, a second gate electrode **330**, a second interlayer insulating layer **340**, a second source electrode **350** and a second drain electrode **360**.

(34) The second semiconductor pattern **310** can be disposed close to the separation insulating layer **130**. For example, the second semiconductor pattern **310** can be in direct contact with the separation insulating layer **130**. The second semiconductor pattern **310** can include a semiconductor. The second semiconductor pattern **310** can include a material different from the first semiconductor pattern **210**. For example, the second semiconductor pattern **310** can include an oxide semiconductor, such as IGZO.

(35) The second semiconductor pattern **310** can include a second source region, a second drain region and a second channel region. The second channel region can be disposed between the second source region and the second drain region. The second source region and the second drain region can have a lower resistance than the second channel region. For example, each of the second source region and the second drain region can be a conductorized region. The second channel region can be a region which is not conductorized.

(36) The second gate insulating layer **320** can be disposed on the second semiconductor pattern **310**. The second gate insulating layer **320** can include an insulating material. For example, the second gate insulating layer **320** can include silicon oxide based material (SiOx), silicon nitride based material (SiNx) and/or a material having a high dielectric constant (High-K material). The second gate insulating layer **320** can have a multi-layer structure.

(37) The second gate insulating layer **320** can expose the second source region and the second drain region of the second semiconductor pattern **310**. The second source region and the second drain region of the second semiconductor pattern **310** may not overlap the second gate insulating layer **320**. For example, the second source region and the second drain region of the second semiconductor pattern **310** can be conductorized by an etchant used in a process of patterning the second gate insulating layer **320**.

(38) The second gate electrode **330** can be disposed on the second gate insulating layer **320**. For example, the second gate electrode **330** can overlap the second channel region of the second semiconductor pattern **310**. The second gate electrode **330** can include a conductive material. For example, the second gate electrode **330** can include a metal, such as aluminum (Al), chromium (Cr), copper (Cu), titanium (Ti), molybdenum (Mo) and tungsten (W). The second gate electrode **330** can include the same material as the first gate electrode **230**.

(39) The second interlayer insulating layer **340** can be disposed on the second semiconductor pattern **310** and the second gate electrode **330**. The second interlayer insulating layer **340** can cover a side surface of the second semiconductor pattern **310**. The second interlayer insulating layer **340** can include an insulating material. The second interlayer insulating layer **340** can include a material

different from the first interlayer insulating layer **240**. For example, the second interlayer insulating layer **340** can include silicon oxide based material (SiOx).

(40) The second source electrode **350** can be disposed on the second interlayer insulating layer **340**. The second source electrode **350** can be electrically connected to the second source region of the second semiconductor pattern **310**. For example, the second interlayer insulating layer **340** can include a second source contact hole partially exposing the second source region of the second semiconductor pattern **310**. The second source electrode **350** can include a portion overlapping with the second source region of the second semiconductor pattern **310**.

(41) The second source electrode **350** can include a conductive material. For example, the second source electrode **350** can include a metal, such as aluminum (Al), chromium (Cr), molybdenum (Mo), tungsten (W) and copper (Cu). The second source electrode **350** can include the same material as the first source electrode **250**.

(42) The second drain electrode **360** can be disposed on the second interlayer insulating layer **340**. The second drain electrode **360** can be electrically connected to the second drain region of the second semiconductor pattern **310**. For example, the second interlayer insulating layer **340** can include a second drain contact hole partially exposing the second drain region of the second semiconductor pattern **310**. The second drain electrode **360** can include a portion overlapping with the second drain region of the second semiconductor pattern **310**.

(43) The second drain electrode **360** can include a conductive material. For example, the second drain electrode **360** can include a metal, such as aluminum (Al), chromium (Cr), molybdenum (Mo), tungsten (W) and copper (Cu). The second drain electrode **360** can include the same material as the second source electrode **350**. For example, the second drain electrode **360** can be formed by the same process as the second source electrode **350**.

(44) The second source electrode **350** and the second drain electrode **360** can have a multi-layer structure. For example, each of the second source electrode **350** and the second drain electrode **360** can have a three-layer in which an intermediate layer being composed of a titanium (Ti) metal layer is disposed between a lower layer and an upper layer which are composed of an aluminum (Al) metal layer.

(45) The storage capacitor **400** can be formed between the device substrate **100** and the second thin film transistor **300**. For example, the storage capacitor **400** can include a first storage electrode **410** disposed on the same layer as the first gate electrode **230**, and a second storage electrode **420** on the first storage electrode **410**.

(46) The first storage electrode **410** can include a conductive material. The first storage electrode **410** can include a metal, such as aluminum (Al), chromium (Cr), molybdenum (Mo), tungsten (W) and copper (Cu). The first storage electrode **410** can include the same material as the first gate electrode **230**. For example, the first storage electrode **410** can be formed by the same process as the first gate electrode **230**.

(47) The second storage electrode **420** can include a conductive material. The second storage electrode **420** can include a metal, such as aluminum (Al), chromium (Cr), molybdenum (Mo), tungsten (W) and copper (Cu). The first interlayer insulating layer **240** can extend between the first storage electrode **410** and the second storage electrode **420**. The second storage electrode **420** can be disposed on the first interlayer insulating layer **240**. The first source electrode **250** and the first drain electrode **260** can be disposed a layer different from the second storage electrode **420**. For example, a first intermediate insulating layer **120** covering the second storage electrode **420** can be disposed between the first interlayer insulating layer **240** and the first source electrodes **250**, and between the first interlayer insulating layer **240** and the first drain electrode **260**. The first interlayer insulating layer **240** and the first intermediate insulating layer **120** can be sequentially stacked between the first gate electrode **230** and the first source electrode **250**, and between the first gate electrode **230** and the first drain electrode **260**. The second storage electrode **420** can include a material different from the first source electrode **250** and the first drain electrode **260**.

(48) The first intermediate insulating layer **120** can include an insulating material. For example, the first intermediate insulating layer **120** can include silicon oxide based material (SiOx) and/or silicon nitride based material (SiNx). The first intermediate insulating layer **120** can have a multi-layer structure. For example, the first intermediate insulating layer **120** can have a stacked structure of a first intermediate layer **121** including silicon oxide based material (SiOx), and a second intermediate layer **122** including silicon nitride based material (SiNx). The first source electrode **250** and the first drain electrode **260** can be disposed on the second intermediate layer **122**. However, the disclosure is not limited thereto. For example, the first intermediate insulating layer **120** can be formed as a single layer of the second intermediate layer **122** including silicon nitride based material (SiNx).
(49) The second storage electrode **420** can be electrically connected to the second drain electrode **360** of the second thin film transistor **300**. For example, a first intermediate electrode **610** connected to the second storage electrode **420** by penetrating the first intermediate insulating layer **120** can be disposed on the second intermediate layer **122**, the second drain electrode **360** can be connected to the first intermediate electrode **610** by penetrating the second interlayer insulating layer **340**. The first intermediate electrode **610** can include a conductive material. For example, the first intermediate electrode **610** can include a metal, such as aluminum (Al), chromium (Cr), molybdenum (Mo), tungsten (W) and copper (Cu). The first intermediate electrode **610** can include the same material as the first source electrode **250** and the first drain electrode **260**. For example, the first intermediate electrode **610** can be formed by the same process as the first source electrode **250** and the first drain electrode **260**.
(50) A light-blocking electrode **450** can be disposed between the first interlayer insulating layer **240** and the first intermediate insulating layer **120**. The light-blocking electrode **450** can overlap the second semiconductor pattern **310**. The light-blocking electrode **450** can prevent changes in characteristics of the second semiconductor pattern **310** due to external light. For example, the light-blocking electrode **450** can include a metal. The light-blocking electrode **450** can include the same material as the second storage electrode **420**. For example, the light-blocking electrode **450** can be formed by the same process as the second storage electrode **420**.
(51) A second intermediate insulating layer **140** can be disposed between the second interlayer insulating layer **340** and the second source and drain electrodes **350** and **360**. The second source electrode **350** and the second drain electrode **360** can be disposed on the second intermediate insulating layer **140**. The second intermediate insulating layer **140** can include an insulating material. The second intermediate insulating layer **140** can include a material different from the second interlayer insulating layer **340**. For example, the second intermediate insulating layer **140** can include silicon nitride based material (SiNx). The second source electrode **350** and the second drain electrode **360** can penetrate the second interlayer insulating layer **340** and the second intermediate insulating layer **140**.
(52) A buffer insulating layer **110** can be disposed between the device substrate **100** and the driving circuit of each pixel PA. The buffer insulating layer **110** can prevent pollution from the device substrate **100** during a process of forming the driving circuits. For example, the buffer insulating layer **110** can extend between the device substrate **100** and the first semiconductor pattern **210** of each pixel PA. The buffer insulating layer **110** can include an insulating material. For example, the buffer insulating layer **110** can include silicon oxide based material (SiOx) and/or silicon nitride based material (SiNx). The buffer insulating layer **110** can have a multi-layer structure. For example, the buffer insulating layer **110** can have a stacked structure of a first buffer layer **111** and a second buffer layer **112** including a material different from the first buffer layer **111**.
(53) A first over-coat layer **150** and a second over-coat layer **160** can be sequentially stacked between the second thin film transistor **300** and the light-emitting device **500** of each pixel PA. The first over-coat layer **150** and the second over-coat layer **160** can remove a thickness difference due to the driving circuit of each pixel PA. For example, a surface of the second over-coat layer **160**

toward the light-emitting device **500** of each pixel PA can be a flat surface. The first over-coat layer **150** and the second over-coat layer **160** can include an insulating material. The first over-coat layer **150** and the second over-coat layer **160** can include a material different from the second intermediate insulating layer **140**. For example, the first over-coat layer **150** and the second over-coat layer **160** can include an organic insulating material. The second over-coat layer **160** can include a material different from the first over-coat layer **150**.
(54) The light-emitting device **500** of each pixel PA can be electrically connected to the second thin film transistor **300** of the corresponding pixel PA. For example, the first electrode **510** of each pixel PA can be electrically connected to the corresponding second drain electrode **360** by penetrating the first over-coat layer **150** and the second over-coat layer **160**. The first electrode **510** of each pixel PA can be electrically connected to the corresponding second drain electrode **360** by second intermediate electrode **620**. For example, the second intermediate electrode **620** can be disposed between the first over-coat layer **150** and the second over-coat layer **160**. In each pixel PA, the second intermediate electrode **620** can be connected to the second drain electrode **360** by penetrating the first over-coat layer **150**, and the first electrode **510** can be connected to the second intermediate electrode **620** by penetrating the second over-coat layer **160**.
(55) The second intermediate electrode **620** can include a conductive material. For example, the second intermediate electrode **620** can include a metal, such as aluminum (Al), chromium (Cr), molybdenum (Mo), tungsten (W) and copper (Cu). The second intermediate electrode **620** can include a material different from the first intermediate electrode **610**.
(56) The light-emitting device **500** of each pixel PA can be independently operated. For example, the first electrode **510** of each pixel PA can be insulated from the first electrode **510** of adjacent pixel PA. An edge of each first electrode **510** can be covered by a bank insulating layer **170**. The bank insulating layer **170** can be disposed on the second over-coat layer **160**. The light-emitting layer **520** and the second electrode **530** of each pixel PA can be stacked on a portion of the corresponding first electrode **510** exposed by the bank insulating layer **170**. The bank insulating layer **170** can include an insulating material. For example, the bank insulating layer **170** can include an organic insulating material. The bank insulating layer **170** can include a material different from the second over-coat layer **160**.
(57) At least a portion of the light-emitting layer **520** of each pixel PA can extend onto the bank insulating layer **170**. For example, the first organic layer **521** and the second organic layer **523** of each pixel PA can be connected to the first organic layer **521** and the second organic layer **523** of adjacent pixel PA. The emission material layer **522** of each pixel PA can be spaced away from the emission material layer **522** of adjacent pixel PA. The second electrode **530** of each pixel PA can extend onto the bank insulating layer **170**. For example, the second electrode **530** of each pixel PA can be connected to the second electrode **530** of adjacent pixel PA.
(58) In the display apparatus according to the embodiment of the present invention, the second thin film transistor **300** of each pixel PA can serve as a driving transistor. Thus, in the display apparatus according to the embodiment of the present invention, the first electrode **510** of each light-emitting device **500** can be connected to the corresponding second thin film transistor **300**. However, the disclosure is not limited thereto. For example, the first electrode **510** of each light-emitting device **500** can be connected to the first thin film transistor **200** of the corresponding pixel PA, and the first thin film transistor **200** of each pixel PA can serve as a driving transistor.
(59) A substrate hole CH can be formed in the device substrate **100**. The substrate hole CH can penetrate the device substrate **100**. The substrate hole CH can be disposed among the pixels PA. For example, the substrate hole CH can be formed among the light-emitting devices **500**. The device substrate **100** can include a hole peripheral region HA including a region in which the substrate hole CH is formed. The gate line GL and the data line DL can bypass the substrate hole CH along an edge of the substrate hole CH in the hole peripheral region HA.
(60) The hole peripheral region HA can include a penetrating region CA in which the substrate hole

CH is formed, and a separating region SA surrounding the penetrating region CA. For example, the separating region SA can be disposed between the penetrating region CA and the pixels PA.
(61) At least one separating device **700** can be disposed in the separating region SA. The separating device **700** can have a structure in which wide patterns **710** and narrow patterns **720** are repeatedly stacked. For example, each of the wide patterns **710** can include a first wide pattern **711**, a second wide pattern **712** and a third wide pattern **713**, which are sequentially stacked, and each of the narrow patterns **720** can include a first narrow pattern **721** between the first wide pattern **711** and the second wide pattern **712**, and a second narrow pattern **722** between the second wide pattern **712** and the third wide pattern **713**. A width w**2** of each narrow pattern **720** can be smaller than a width w**1** of the first wide pattern **711**. Each of the separating devices **700** can include at least one under-cut structure due to a width difference of the wide patterns **710** and the narrow patterns **720**.
(62) The under-cut structure can have a depth h (or height h) that is the same as a thickness of the corresponding narrow pattern **720**. For example, the depth h of the under-cut structure formed by the third wide pattern **713** and the second narrow pattern **722** can be the same as the thickness of the second narrow pattern **722**. A length d (or width d) of the under-cut structure can be the same as a distance between a side surface of the corresponding wide pattern **710** and a side surface of the corresponding narrow pattern **720**. For example, the length d of the under-cut structure formed by the third wide pattern **713** and the second narrow pattern **722** can be the same as the distance between a side surface of the third wide pattern **713** and a side surface of the second narrow pattern **722**. The depth h and the length d of the under-cut structure can be larger than a thickness of the light-emitting layer **520** which extends on the hole peripheral region HA. For example, the depth h and the length d of the under-cut structure can be larger than a sum of a thickness of the first organic layer **521** and a thickness of the second organic layer **523**. For example, the depth h and the length d of the under-cut structure can be larger than 3000 μm, respectively. Thus, in the display apparatus according to the embodiment of the present invention, the light-emitting layer **520** and the second electrode **530** which are deposited on the separating region SA can be completely separated by the separating device **700**. For example, the separating device **700** can separate the first organic layer **521** and the second organic layer **523** of the light-emitting layer **520** and the second electrode **530** in the separating region SA. Therefore, in the display apparatus according to the embodiment of the present invention, the permeation of the external moisture via the light-emitting layer **520** can be blocked by the separating device **700**. The second electrode **530** can be separated by the under-cut structure of the separating device **700**. For example, in the display apparatus according to the embodiment of the present invention, the under-cut structure of the separating device **700** can be exposed by the light-emitting layer **520** and the second electrode **530**.
(63) The wide patterns **710** and the narrow patterns **720** can include an insulating material. The narrow patterns **720** can include a material different from the wide patterns **710**. For example, the narrow patterns **720** can include a material that is etched faster than the wide patterns **710** by an etchant used in a process of forming the separating device **700**. Thus, in the display apparatus according to the embodiment of the present invention, the process of forming the separating device **700** can be simplified.
(64) The first wide pattern **711** can be spaced away from a portion of the light-emitting layer **520** and a portion of the second electrode **530**, which are separated by the separating device **700**. For example, a width w**1** of the first wide pattern **711** can be smaller than a width of the third wide pattern **713** which is disposed at the uppermost portion of the separating device **700**.
(65) The separating device **700** can be formed by a process of forming the driving circuit and the light-emitting device **500**. For example, a step of forming the separating device **700** can include a step of depositing the first interlayer insulating layer **240**, the first intermediate layer **121**, the second intermediate layer **122**, the separation insulating layer **130**, the second interlayer insulating layer **340**, the second intermediate insulating layer **140** and the second over-coat layer **160** on the separating region SA, and a step of sequentially etching the first intermediate layer **121**, the second

intermediate layer **122**, the separation insulating layer **130**, the second interlayer insulating layer **340**, the second intermediate insulating layer **140** and the second over-coat layer **160** using a mask pattern. The first wide pattern **711** can be formed by a process of etching the first intermediate layer **121** on the separating region SA. For example, the first wide pattern **711** can include silicon oxide based material (SiOx). The first narrow pattern **721** can be formed by a process of etching the second intermediate layer **122** on the separating region SA. For example, the first narrow pattern **721** can include silicon nitride based material (SiNx). The second wide pattern **712** can be formed by a process of etching the separation insulating layer **130** and the second interlayer insulating layer **340** on the separating region SA. For example, the second wide pattern **712** can include silicon oxide based material (SiOx). The second wide pattern **712** can have a stacked structure of a lower wide layer **712***a* and an upper wide layer **712***b*. The second narrow pattern **722** can be formed by a process of etching the second intermediate insulating layer **140** on the separating region SA. For example, the second narrow pattern **722** can include silicon nitride based material (SiNx). The third wide pattern **713** can be formed by a process of etching the second over-coat layer **160**. For example, the third wide pattern **713** can include an organic insulating material. Thus, in the display apparatus according to the embodiment of the present invention, decreasing the process efficiency due to the process of forming the separating device **700** can be minimized.
(66) The hole peripheral region HA can further include a barrier region BA disposed outside the separating region SA. The separating region SA can be disposed between the penetrating region CA and the barrier region BA. At least one dam **800** can be disposed on the barrier region BA.
(67) Accordingly, the display apparatus according to the embodiment of the present invention includes at least one separating device **700** within the separating region SA disposed between the penetrating region CA in which the substrate hole CH is formed, and the pixels PA, wherein the separating device **700** can include at least one under-cut structure, and wherein the depth and the length of the under-cut structure can larger than the thickness of the light-emitting layer **520** of the light-emitting device **500** in each pixel PA. Thus, in the display apparatus according to the embodiment of the present invention, the light-emitting layer **520** can be completely separated by the separating device **700**. For example, in the display apparatus according to the embodiment of the present invention, the damage of the light-emitting devices **500** due to the external moisture permeating through the substrate hole CH can be effectively prevented. And, in the display apparatus according to the embodiment of the present invention, the separating device **700** can be formed using the process of forming the driving circuit and the light-emitting device **500** of each pixel PA. Therefore, in the display apparatus according to the embodiment of the present invention, decreasing the process efficiency due to the process of forming the separating device **700** can be prevented.
(68) The display apparatus according to the embodiment of the present invention is described that each of the device substrate **100** and the buffer insulating layer **110** has a multi-layer structure. However, other variations and examples of the display apparatus according to other embodiments of the present disclosure are possible as shown in FIGS. **6** to **15**.
(69) Particularly, FIGS. **6** to **15**B are views respectively showing a display apparatus according to other embodiments of the present invention, where FIG. **7**B is an enlarged view of an area P**2** in FIG. **7**A, FIG. **8**B is an enlarged view of an area P**3** in FIG. **8**A, FIG. **10**B is an enlarged view of an area P**4** in FIG. **10**A, FIG. **12**B is an enlarged view of an area P**5** in FIG. **12**A, FIG. **14**B is an enlarged view of an area P**6** in FIG. **14**A, and FIG. **15**B is an enlarged view of an area P**7** in FIG. **15**A.
(70) For example, the display apparatus according to another embodiment of the present invention can include the device substrate **100** having a single layer structure, and the buffer insulating layer **110** having a single layer structure, as shown in FIG. **6**. Alternatively, in the display apparatus according to another embodiment of the present invention, only one of the device substrate **100** and the buffer insulating layer **110** can have a single layer structure.

(71) The display apparatus according to the embodiment of the present invention is described that the second intermediate insulating layer **140** is disposed between the second interlayer insulating layer **340** and the first over-coat layer **150**. However, in the display apparatus according to another embodiment of the present invention, the second thin film transistor **300** can include a passivation layer **345** insulating the second gate electrode **330** from the second source electrode **350** and the second drain electrode **360**, as shown in FIGS. **6**, **7**A and **7**B. For example, the passivation layer **345** can be disposed between the second gate electrode **330** and the second source electrode **350**, and between the second gate electrode **330** and the second drain electrode **360**. The passivation layer **345** can be in direct contact with the second gate electrode **330**, the second source electrode **350** and the second drain electrode **360**. The passivation layer **345** can include silicon oxide based material (SiOx). For example, in the display apparatus according to another embodiment of the present invention, an insulating layer including silicon nitride based material (SiNx) may not be disposed on the second semiconductor pattern **310** of the second thin film transistor **300**. Thus, in the display apparatus according to another embodiment of the present invention, the deterioration of the second semiconductor pattern **310** including an oxide semiconductor due to hydrogen generated from an insulating layer including silicon nitride based material (SiNx) can be prevented.
(72) The display apparatus according to the embodiment of the present invention is described that the wide patterns **710** and the narrow patterns **720** include an insulating material. However, in the display apparatus according to another embodiment of the present invention, the wide patterns **710** and/or the narrow patterns **720** can include a conductive material. For example, in the display apparatus according to another embodiment of the present invention, the separating device **700** can be disposed on the passivation layer **345**, as shown in FIGS. **6**, **7**A and **7**B. The separating device **700** can include a single wide pattern **710** and a single narrow pattern **720**. For example, the separating device **700** can include the narrow pattern **720** and the wide pattern **710** which are sequentially stacked on the passivation layer **345**.
(73) The narrow pattern **720** can include a conductive material. The narrow pattern **720** can be disposed on the same layer as the second source electrode **350** and the second drain electrode **360** of the second thin film transistor **300**. For example, the narrow pattern **720** can include the same material as the second source electrode **350** and the second drain electrode **360** of the second thin film transistor **300**. The narrow pattern **720** can be formed by the same process as the second source electrode **350** and the second drain electrode **360** of the second thin film transistor **300**. The narrow pattern **720** can have the same structure as the second source electrode **350** and the second drain electrode **360** of the second thin film transistor **300**. For example, the narrow pattern **720** can have a three-layer structure in which an intermediate layer being composed of a titanium (Ti) metal layer is disposed between a lower layer and an upper layer which are composed of an aluminum (Al) metal layer. The wide pattern **710** can include the same material as one of the first over-coat layer **150**, the second over-coat layer **160** and the bank insulating layer **170**, which are disposed on the second source electrode **350** and the second drain electrode **360** of the second thin film transistor **300**. Thus, in the display apparatus according to another embodiment of the present invention, the separating device **700** can be effectively formed. For example, in the display apparatus according to another embodiment of the present invention, the depth and the length of the under-cut structure due to the wide pattern **710** and the narrow pattern **720** can be easily controlled. Therefore, in the display apparatus according to another embodiment of the present invention, the separation of the second electrode **530** and the light-emitting layer **520** by the separating device **700** can be effectively performed. For example, the first organic layer **521** and the second organic layer **523** of the light-emitting layer **520** and the second electrode **530** can be separated in a plurality of patterns on the separating region SA by the separating device **700**, as shown in FIG. **7**A.
(74) The display apparatus according to the embodiment of the present invention is described that the narrow patterns **720** includes the same material. However, in the display apparatus according to

another embodiment of the present invention, the narrow patterns **720** can have a stacked structure of layers formed of different materials. For example, in the display apparatus according to another embodiment of the present invention, the separating device **700** can include a first wide pattern **711**, a first narrow pattern **721**, a second wide pattern **712**, a second narrow pattern **722** and a third wide pattern **713**, which are sequentially stacked on the first interlayer insulating layer **240**, as shown in FIGS. **6**, **8**A and **8**B. The first wide pattern **711** and the second wide pattern **712** can include silicon oxide based material (SiOx). The first wide pattern **711** can include the same material as the first intermediate layer **121** of the first intermediate insulating layer **120**. The first wide pattern **711** can be disposed on the same layer as the first intermediate layer **121** of the first intermediate insulating layer **120**.
(75) The second wide pattern **712** can have a two-layer structure in which a lower wide layer **712***a* and an upper wide layer **712***b* are stacked. The lower wide layer **712***a* of the second wide pattern **712** can include the same material as the separation insulating layer **130**. The lower wide layer **712***a* of the second wide pattern **712** can be disposed on the same layer as the separation insulating layer **130**. The upper wide layer **712***b* of the second wide pattern **712** can include the same material as the passivation layer **345**. The upper wide layer **712***b* of the second wide pattern **712** can be disposed on the same layer as the passivation layer **345**.
(76) The third wide pattern **713** can include the same material as one of the first over-coat layer **150**, the second over-coat layer **160** and the bank insulating layer **170** which are disposed on the second source electrode **350** and the second drain electrode **360** of the second thin film transistor **300**. For example, the third wide pattern **713** can include the same material as the second over-coat layer **160**.
(77) The first narrow pattern **721** can include the same material as the second intermediate layer **122** of the first intermediate insulating layer **120**. For example, the first narrow pattern **721** can include silicon oxide based material (SiOx). The first narrow pattern **721** can be disposed on the same layer as the second intermediate layer **122** of the first intermediate insulating layer **120**. The second narrow pattern **722** can include the same material as the second source electrode **350** and the second drain electrode **360** of the second thin film transistor **300**. For example, the second narrow pattern **722** can be formed in the same structure as the second source electrode **350** and the second drain electrode **360**. The second narrow pattern **722** can have a three-layer structure in which an intermediate layer being composed of a titanium (Ti) metal layer is disposed between a lower layer and an upper layer which are composed of an aluminum (Al) metal layer. Thus, in the display apparatus according to another embodiment of the present invention, the degree of freedom for configuration of the second narrow pattern **722** can be improved.
(78) The display apparatus according to the embodiment of the present invention is described that the second thin film transistor **300** of each pixel PA is electrically connected to the first electrode **510** of the corresponding pixel PA. However, in the display apparatus according to another embodiment of the present invention, the first electrode **510** of each pixel PA can be electrically connected to the first thin film transistor **200** of the corresponding pixel PA, as shown in FIG. **9**. For example, the display apparatus according to another embodiment of the present invention can include a third intermediate electrode **630** connected to the first drain electrode **260** of the first thin film transistor **200** by penetrating the separation insulating layer **130**, the second interlayer insulating layer **340** and the second intermediate insulating layer **140**, and a fourth intermediate electrode **640** connected to the third intermediate electrode **630** by penetrating the first over-coat layer **150**. The first electrode **510** of each pixel PA can be connected to the fourth intermediate electrode **640** by penetrating the second over-coat layer **160**. Thus, in the display apparatus according to another embodiment of the present invention, the degree of freedom for the location of each driving circuit can be improved.
(79) The display apparatus according to the embodiment of the present invention is described that the second thin film transistor **300** including an oxide semiconductor is electrically connected to the

first electrode **510** of the light-emitting device **500** to serve as a driving transistor, as shown in FIGS. **3** and **6**. However, the display apparatus according to another embodiment of the present invention can include the first thin film transistor **200** including a poly silicon electrically connected to the first electrode **510** of the light-emitting device **500** to serve as a driving transistor, as shown in FIGS. **9**, **10**A and **10**B. In the display apparatus according to another embodiment of the present invention, the first intermediate insulating layer **120** can be a single layer including silicon nitride based material (SiNx). However, the disclosure is not limited thereto. For example, the first intermediate insulating layer **120** can have a two-layer structure including a layer composed of silicon nitride based material (SiNx), and a layer composed of silicon oxide based material (SiOx).

(80) In the display apparatus according to another embodiment of the present invention, the separating device **700** can include a first wide pattern **711**, a first narrow pattern **721**, a second wide pattern **712**, a second narrow pattern **722** and a third wide pattern **713**, which are sequentially stacked on the second buffer layer **112**. The first wide pattern **711** and the second wide pattern **712** can include silicon oxide based material (SiOx). For example, the first wide pattern **711** can include the same material as the first gate insulating layer **220**, and the second wide pattern **712** can have a stacked structure of a lower wide layer **712***a* including the same material as the separation insulating layer **130**, and an upper wide layer **712***b* including the same material as the second interlayer insulating layer **340**. The third wide pattern **713** can include the same material as one of the first over-coat layer **150**, the second over-coat layer **160** and the bank insulating layer **170**, which are disposed on the second source electrode **350** and the second drain electrode **360** of the second thin film transistor **300**. For example, the third wide pattern **713** can be formed by a mask process of forming the second over-coat layer **160**. The third wide pattern **713** can include the same material as the second over-coat layer **160**. The first narrow pattern **721** and the second narrow pattern **722** can include silicon nitride based material (SiNx). For example, the first narrow pattern **721** can have a stacked structure of a lower narrow layer **721***a* including the same material as the first interlayer insulating layer **240**, and an upper narrow layer **721***b* including the same material as the first intermediate insulating layer **120**. The second narrow pattern **722** can include the same material as the second intermediate insulating layer **140**. Thus, in the display apparatus according to another embodiment of the present invention, the degree of freedom for the configuration of the separating device **700** can be improved.

(81) In the display apparatus according to another embodiment of the present invention, the second gate electrode **330** can be insulated from the second source electrode **350** and the second drain electrode **360** by the passivation layer **345**, as shown in FIGS. **11**, **12**A and **12**B. For example, in the display apparatus according to another embodiment of the present invention, the separating device **700** can include the first wide pattern **711**, the first narrow pattern **721**, the second wide pattern **712**, the second narrow pattern **722** and the third wide pattern **713**, which are sequentially stacked on the buffer insulating layer **110**. The first wide pattern **711** can include the same material as the first gate insulating layer **220**. The second wide pattern **712** can have a stacked structure of a lower wide layer **712***a* including the same material as the separation insulating layer **130**, and an upper wide layer **712***b* including the same material as the passivation layer **345**. The third wide pattern **713** can include the same material as one of the first over-coat layer **150**, the second over-coat layer **160** and the bank insulating layer **170**, which are disposed on the second source electrode **350** and the second drain electrode **360** of the second thin film transistor **300**. For example, the third wide pattern **713** can be formed by a mask process of forming the second over-coat layer **160**. The third wide pattern **713** can include the same material as the second over-coat layer **160**. The first narrow pattern **721** can have a stacked structure of a lower narrow layer **721***a* including the same material as the first interlayer insulating layer **240**, and an upper narrow layer **721***b* including the same material as the first intermediate insulating layer **120**. The second narrow pattern **722** can include the same material as the second source electrode **350** and the second drain electrode **360** of

the second thin film transistor **300**. Thus, in the display apparatus according to another embodiment of the present invention, the separating device **700** can be effectively configured. The second narrow pattern **722** can have a three-layer structure composed of a lower layer, an intermediate layer and an upper layer similarly to the second source electrode **350** and the second drain electrode **360**. For example, the lower layer and the upper layer of the second narrow pattern **722** can be composed of an aluminum (Al) metal layer, and the intermediate layer of the second narrow pattern **722** between the lower layer and the upper layer of the second narrow pattern **722** can be composed of a titanium (Ti) metal layer.

(82) In the display apparatus according to another embodiment of the present invention, the second gate electrode **330** can be insulated from the second source electrode **350** and the second drain electrode **360** by the passivation layer **345**, and the separating device **700** can include a single narrow pattern **720** and a single wide pattern **710**, which are sequentially stacked on the passivation layer **345**, as shown in FIGS. **11** and **13**.

(83) The narrow pattern **720** can include a conductive material. For example, the narrow pattern **720** can include the same material as the second source electrode **350** and the second drain electrode **360** of the second thin film transistor **300**. The narrow pattern **720** can be disposed on the same layer as the second source electrode **350** and the second drain electrode **360** of the second thin film transistor **300**. The narrow pattern **720** can be formed by the same process as the second source electrode **350** and the second drain electrode **360** of the second thin film transistor **300**. For example, the second source electrode **350**, the second drain electrode **360** and the narrow pattern **720** can have a structure in which a titanium (Ti) metal layer is disposed between aluminum (Al) metal layers. When the narrow pattern **720** includes the same material as the second source electrode **350** and the second drain electrode **360**, the wide pattern **710** can include the same material as one of the first over-coat layer **150**, the second over-coat layer **160** and the bank insulating layer **170**. However, the disclosure is not limited thereto. For example, the narrow pattern **720** can include the same material as the fourth intermediate electrode **640** between the first over-coat layer **150** and the second over-coat layer **160**. The narrow pattern **720** can be formed by the same process as the fourth intermediate electrode **640**. When the narrow pattern **720** includes the same material as the fourth intermediate electrode **640**, the wide pattern **710** can include the same material as one of the second over-coat layer **160** and the bank insulating layer **170**. Thus, in the display apparatus according to another embodiment of the present invention, the under-cut structure can be formed by the wide pattern **710** and the narrow pattern **720**, and the separation of the second electrode **530** and the light-emitting layer **520** can be effectively performed. For example, the first organic layer **521** and the second organic layer **523** of the light-emitting layer **520** and the second electrode **530** can be separated in a plurality of patterns on the separating region SA by the separating device **700**, as shown in FIG. **13**.

(84) In the display apparatus according to another embodiment of the present invention, the separating device **700** can include the first wide pattern **731**, the first narrow pattern **741**, the second wide pattern **732**, the second narrow pattern **742** and the third wide pattern **733**, which are sequentially stacked on the passivation layer **345**, as shown in FIGS. **14**A and **14**B. The first wide pattern **731** can include an organic insulating material same as the first over-coat layer **150**. The first narrow pattern **741** can include a metal material same as the fourth intermediate electrode **640**. The second wide pattern **732** can include an organic insulating material same as the second over-coat layer **160**. The second narrow pattern **742** can include a metal material same as the first electrode **510**. The third wide pattern **733** can include an organic insulating material same as the bank insulating layer **170**. Thus, in the display apparatus according to another embodiment of the present invention, the wide patterns **731**, **732** and **733** of the separating device **700** can be composed of an organic material layer, and the narrow patterns **741** and **742** of the separating device **700** can be composed of a metal material layer.

(85) And, in the display apparatus according to another embodiment of the present invention, the

separating device **700** can include the first wide pattern **751**, the first narrow pattern **761**, the second wide pattern **752**, the second narrow pattern **762**, the third wide pattern **753**, the third narrow pattern **763** and the fourth wide pattern **754**, which are sequentially stacked on the first gate insulating layer **220**, as shown in FIGS. **15**A and **15**B. The first wide pattern **751** can include the same material as the first gate electrode **230** of the first thin film transistor **200**. The second wide pattern **752** can include the same material as the first source electrode **250** and the first drain electrode **260** of the first thin film transistor **200**. The third wide pattern **753** can include the same material as the second gate electrode **330** of the second thin film transistor **300**. The fourth wide pattern **754** can include the same material as one of the first over-coat layer **150**, the second over-coat layer **160** and the bank insulating layer **170**. The first narrow pattern **761**, the second narrow pattern **762** and the third narrow pattern **763** can include an insulating material. For example, the first narrow pattern **761** can have a stacked structure of a layer including the same material as the first interlayer insulating layer **240**, and a layer including the same material as the first intermediate insulating layer **120**. The first narrow pattern **761** can have the same thickness as a sum of a thickness of the first interlayer insulating layer **240** and a thickness of the first intermediate insulating layer **120**. The second narrow pattern **762** can include the same material as the separation insulating layer **130** composed of silicon oxide based material (SiOx). The second narrow pattern **762** can be formed by the same mask process as the separation insulating layer **130**. The third narrow pattern **763** can include the same material as the passivation layer **345** composed of silicon oxide based material (SiOx). The third narrow pattern **763** can be formed by the same mask process as the passivation layer **345**. Thus, in the display apparatus according to another embodiment of the present invention, the degree of freedom for the configuration of the separating device **700** can be improved.

(86) As a result, the display apparatus according to the embodiments of the present invention can include at least one separating device between the substrate hole and the light-emitting devices, wherein each of the light-emitting devices can include the light-emitting layer between the first electrode and the second electrode, wherein the separating device can include at least one under-cut structure, and wherein the under-cut structure can have the depth and the length greater than the thickness of the light-emitting layer. Thus, in the display apparatus according to the embodiments of the present invention, the moving path of the external moisture permeating through the substrate hole can be blocked by the separating device. Thereby, in the display apparatus according to the embodiments of the present invention, the lifetime and the reliability of the light-emitting devices can be improved.

Claims

1. A display apparatus comprising: a flexible substrate including a penetrating hole and a pixel area spaced apart from the penetrating hole; a first thin film transistor on the flexible substrate, the first thin film transistor including a first semiconductor layer disposed on a first region of the pixel area; a second thin film transistor spaced apart from the first thin film transistor, the second thin film transistor including a second semiconductor layer disposed on a second region of the pixel area; a conductive pattern disposed between the flexible substrate and the second semiconductor layer, the conductive pattern overlapping with the second region of the pixel area; a first planarization layer on the first thin film transistor and the second thin film transistor; a storage capacitor disposed between the flexible substrate and the first planarization layer, the storage capacitor being disposed on a third region of the pixel area, such that the storage capacitor is spaced apart from the first thin film transistor and the second thin film transistor; a connection electrode on the first planarization layer, the connection electrode being electrically connected to the first thin film transistor or the second thin film transistor; a second planarization layer on the first planarization layer, the second planarization layer covering the connection electrode; a first electrode on the second planarization

layer, the first electrode being electrically connected to the connection electrode; a bank layer on the second planarization layer, the bank layer exposing a portion of the first electrode; a light-emitting layer on the portion of the first electrode exposed by the bank layer, the light-emitting layer including an emission material layer between a first organic layer and a second organic layer; and a second electrode on the light-emitting layer, wherein the storage capacitor includes a capacitor electrode on the third region of the pixel area, wherein the capacitor electrode is disposed on a same layer as the conductive pattern, wherein the second semiconductor layer includes a material different from the first semiconductor layer, and wherein at least one of the first organic layer, the emission material layer and the second organic layer extends to the penetrating hole.
2. The display apparatus according to claim 1, wherein at least one of the first organic layer and the second organic layer extends to the penetrating hole.
3. The display apparatus according to claim 1, wherein the first semiconductor layer includes a low-temperature poly-silicon (LTPS), and the second semiconductor layer includes an oxide semiconductor.
4. The display apparatus according to claim 1, further comprising: a gate line on the substrate; a data line intersecting the gate line, wherein the first electrode is disposed on a region defined by the gate line and the data line, and wherein the gate line and the data line extend along an edge of the penetrating hole.
5. The display apparatus according to claim 1, further comprising a separating device between the first electrode and the penetrating hole.
6. The display apparatus according to claim 5, wherein the separating device includes a plurality of wide patterns and at least one narrow pattern between the plurality of the wide patterns, and wherein the plurality of the wide patterns includes an organic material.
7. The display apparatus according to claim 6, wherein the at least one narrow pattern includes a conductive material.
8. The display apparatus according to claim 6, further comprising a metal pattern between the first electrode and the separating device and between the separating device and the penetrating hole, wherein the metal pattern is spaced away from the separating device.
9. The display apparatus according to claim 6, wherein the separating device includes a metal layer on the plurality of the wide patterns.
10. The display apparatus according to claim 7, wherein the metal layer includes a same material as the second electrode.
11. A display apparatus comprising: a substrate including a pixel area and a penetrating area spaced apart from the pixel area; a first thin film transistor on the pixel area, the first thin film transistor including a first semiconductor layer on a first region of the pixel area; a second thin film transistor spaced apart from the first thin film transistor on the pixel area, the second thin film transistor including a second semiconductor layer on a second region of the pixel area, the second semiconductor layer having a different material from the first semiconductor layer; a conductive pattern disposed between the substrate and the second semiconductor layer, the conductive pattern overlapping with the second region of the pixel area; a first planarization layer on the first thin film transistor and the second thin film transistor; a storage capacitor disposed between the substrate and the first planarization layer, the storage capacitor being disposed on a third region of the pixel area, such that the storage capacitor is spaced apart from the first thin film transistor and the second thin film transistor; a connection electrode on the first planarization layer of the pixel area, the connection electrode electrically connected to the first thin film transistor or the second thin film transistor; a second planarization layer on the first planarization layer, the second planarization layer covering the connection electrode; a first electrode on the second planarization layer, the first electrode electrically connected to the connection electrode; a bank insulating layer on the second planarization layer, the bank insulating layer covering an edge of the first electrode; a light-emitting layer on a portion of the first electrode exposed by the bank insulating layer, the light-emitting

layer including an emission material layer between a first organic layer and a second organic layer; a second electrode on the light-emitting layer; and a penetrating hole within the penetrating area of the substrate, wherein the storage capacitor includes a capacitor electrode on the third region of the pixel area, wherein the capacitor electrode is disposed on a same layer as the conductive pattern, wherein the first planarization layer and the second planarization layer are spaced away from the penetrating area, and wherein at least one of the first organic layer, the emission material layer and the second organic layer includes a region penetrated by the penetrating hole.

12. The display apparatus according to claim 11, wherein the penetrating hole penetrates the penetrating area of the substrate and the second electrode within the penetrating area.

13. The display apparatus according to claim 11, further comprising a separating device between the pixel area and the penetrating hole, wherein the at least one of the first organic layer, the emission material layer and the second organic layer including the region penetrated by the penetrating hole includes a region separated by the separating device.

14. The display apparatus according to claim 13, wherein the separating device is spaced apart from the first planarization layer and the second planarization layer.

15. The display apparatus according to claim 13, wherein the separating device includes a first wide pattern, a second wide pattern on the first wide pattern and a narrow pattern between the first wide pattern and the second wide pattern, wherein a width of the narrow pattern is smaller than a width of the first wide pattern, and wherein the width of the first wide pattern is smaller than a width of the second wide pattern.

16. The display apparatus according to claim 15, wherein at least one of a depth of the narrow pattern and a difference in length between the narrow pattern and the second wide pattern is larger than a thickness of the at least one of the first organic layer, emission material layer and the second organic layer including the region penetrated by the penetrating hole.

17. The display apparatus according to claim 11, wherein the narrow pattern includes a material different from the first wide pattern and the second wide pattern.